confidence_performance_perception_uncertainty_score_SCR

August 25, 2026

1 Multilevel Analysis for Retrospective Confidence based on Skin Conductance Response, Perceived Performance Perception, and Team Gameplay Score

by Elvis Ortega-Ochoa

[1]:
```
import sys
import subprocess

subprocess.check_call([sys.executable, "-m", "pip", "install", "--upgrade",␣
 ↪"pip", "ipywidgets", "jupyterlab_widgets"])
subprocess.check_call([
    sys.executable, "-m", "pip", "install",
    "pymer4",
    "rpy2",
    "polars",
    "great-tables",
    "joblib",
    "scikit-learn",
    "pyarrow",
    "bambi",
    "arviz",
    "matplotlib",
    "pandas",
    "numpy",
    "statsmodels",
    "openpyxl",
])
```

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Requirement already satisfied: pip in
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-packages
(26.2.1)
Requirement already satisfied: ipywidgets in
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(8.1.9)
Requirement already satisfied: jupyterlab_widgets in
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(3.0.17)
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(from ipywidgets) (0.2.3)
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Requirement already satisfied: pymer4 in
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Requirement already satisfied: arviz_plots<1,>=0.8.0 in
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(from bambi) (0.21)
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(from arviz_plots<1,>=0.8.0->bambi) (0.8.2)
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(from arviz) (1.8.1)
Requirement already satisfied: h5py in
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(from arviz) (3.16.0)
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Requirement already satisfied: filelock>=3.15 in
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(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (3.25.2)
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(from pytensor<2.39,>=2.38.0->pymc<6,>=5.27.0->bambi) (0.4.7)
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(from matplotlib) (1.3.3)
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(from importlib-metadata->great-tables) (4.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in
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(from jinja2->rpy2-robjects>=3.6.5->rpy2) (3.0.3)
```

[1]:
```
0
```

Packages

[2]:
```
import pandas as pd
import numpy as np
import bambi as bmb
import arviz as az
import matplotlib.pyplot as plt
import matplotlib.patches as patches
import statsmodels.api as sm
import pymer4.models as pm
import polars as pl
import ipywidgets as widgets
from pymer4.models import lmer as Lmer
```

Loading data

[3]:
```
# Reading files and printing columns
scr = pd.read_excel("../../confidence_SCR/stable_scr_events_UPD.xlsx",␣
 ↪keep_default_na=False)
cf = pd.read_excel("../../confidence_SCR/CONT_dependent_variables_confidence_E.
 ↪xlsx", keep_default_na=False)

# Make a clean copy
scr = scr.copy()
cf = cf.copy()

# Extract participantID and sensorID from scrID
scr["participantID"] = scr["scrID"].str.extract(r"SCR_([^.]+)\.")
scr["sensorID"] = scr["scrID"].str.extract(r"\.([A-Z]{2})-EDA$")

# Merge the dataframes on groupID, participantID, and sensorID
dataset = scr.merge(
    cf,
    on=["groupID", "participantID", "sensorID"],
    how="inner"
)

# Rename columns for clarity
dataset = dataset.rename(columns={
    "realGamePeaks/m": "realGamePeaks_per_m",
    "realTutoPeaks/m": "realTutoPeaks_per_m",
    "realQuestionsPeaks/m": "realQuestionsPeaks_per_m",
    "realPitchPeaks/m": "realPitchPeaks_per_m",
    "CF1": "cf",
    "PP1": "pp"
})
```

Build questionPath as numeric

[4]:
```
# Build numeric questionPath at group level
dataset["qBonusPeaks"] = pd.to_numeric(dataset["qBonusPeaks"], errors="coerce")
dataset["qPenaltyPeaks"] = pd.to_numeric(dataset["qPenaltyPeaks"],␣
 ↪errors="coerce")

dataset["has_bonus"] = dataset["qBonusPeaks"].notna()
dataset["has_penalty"] = dataset["qPenaltyPeaks"].notna()

group_path = (
    dataset.groupby("groupID")[["has_bonus", "has_penalty"]]
    .any()
    .reset_index()
)
```

```
group_path["questionPath"] = np.select(
    [
        group_path["has_bonus"] & ~group_path["has_penalty"],
        group_path["has_penalty"] & ~group_path["has_bonus"],
    ],
    [1, 0],
    default=np.nan,
)

# Drop any existing questionPath variants to avoid suffixes, then merge
dataset = dataset.drop(columns=[c for c in dataset.columns if c.
 ↪startswith("questionPath")], errors="ignore")
dataset = dataset.merge(group_path[["groupID", "questionPath"]], on="groupID",␣
 ↪how="left")
```

Reverse team gameplay score

[5]:
```
# Build numeric uncertScore
dataset["uncertScore"] = pd.to_numeric(dataset["uncertScore"], errors="coerce")

# Reverse-coding, using the instrument's theoretical bounds, not the sample's
SCALE_MIN, SCALE_MAX = 1, 70  # instrument range, per ¶102
dataset["uncertScore"] = (SCALE_MIN + SCALE_MAX) - dataset["uncertScore"]

# Reverse-score uncertScore using the observed scale range
#dataset["uncertScore"] = dataset["uncertScore"].max() + dataset["uncertScore"].
 ↪min() - dataset["uncertScore"]
```

[6]:
```
# Finalize data types
dataset["groupID"] = dataset["groupID"].astype("category")
dataset["cf"] = pd.to_numeric(dataset["cf"], errors="coerce")
dataset["questionPath"] = pd.Categorical(dataset["questionPath"],␣
 ↪categories=[0,1])
```

[7]:
```
# Keep just variables needed
dat_model = dataset[["groupID", "participantID", "cf", "pp", "uncertScore",␣
 ↪"realGamePeaks_per_m", "realTutoPeaks_per_m", "realQuestionsPeaks_per_m",␣
 ↪"realPitchPeaks_per_m", "questionPath", "pitchRole"]].copy()
```

1.1 Mixed Linear Models

1.1.1 Within/Between Descomposition

[8]:
```
def add_within_between(df, raw_col, prefix):
    group_mean = df.groupby("groupID")[raw_col].transform("mean")
    within = df[raw_col] - group_mean
    between = group_mean - group_mean.mean()
```

```
    df[f"{prefix}_group_mean"] = group_mean
    df[f"{prefix}_within"] = within
    df[f"{prefix}_between"] = between

    return df

dat_model = add_within_between(dat_model, "realGamePeaks_per_m", "scr")
dat_model = add_within_between(dat_model, "realTutoPeaks_per_m", "tuto")
dat_model = add_within_between(dat_model, "realQuestionsPeaks_per_m",␣
 ↪"questions")
dat_model = add_within_between(dat_model, "realPitchPeaks_per_m", "pitch")
dat_model = add_within_between(dat_model, "pp", "pp")
```

1.1.2 Standarize predictor variables

[9]:
```
dat_model["realGamePeaks_per_m_z"] = (
    dat_model["realGamePeaks_per_m"] - dat_model["realGamePeaks_per_m"].mean()
) / dat_model["realGamePeaks_per_m"].std()

dat_model["scr_within_z"] = (
    dat_model["scr_within"] - dat_model["scr_within"].mean()
) / dat_model["scr_within"].std()

dat_model["scr_between_z"] = (
    dat_model["scr_between"] - dat_model["scr_between"].mean()
) / dat_model["scr_between"].std()

dat_model["tuto_within_z"] = (
    dat_model["tuto_within"] - dat_model["tuto_within"].mean()
) / dat_model["tuto_within"].std()

dat_model["tuto_between_z"] = (
    dat_model["tuto_between"] - dat_model["tuto_between"].mean()
) / dat_model["tuto_between"].std()

dat_model["questions_within_z"] = (
    dat_model["questions_within"] - dat_model["questions_within"].mean()
) / dat_model["questions_within"].std()

dat_model["questions_between_z"] = (
    dat_model["questions_between"] - dat_model["questions_between"].mean()
) / dat_model["questions_between"].std()

dat_model["pitch_within_z"] = (
    dat_model["pitch_within"] - dat_model["pitch_within"].mean()
) / dat_model["pitch_within"].std()
```

```
dat_model["pitch_between_z"] = (
    dat_model["pitch_between"] - dat_model["pitch_between"].mean()
) / dat_model["pitch_between"].std()

dat_model["pp_z"] = (
    dat_model["pp"] - dat_model["pp"].mean()
) / dat_model["pp"].std()

dat_model["pp_within_z"] = (
    dat_model["pp_within"] - dat_model["pp_within"].mean()
) / dat_model["pp_within"].std()

dat_model["pp_between_z"] = (
    dat_model["pp_between"] - dat_model["pp_between"].mean()
) / dat_model["pp_between"].std()

dat_model["uncertScore_z"] = (
    dat_model["uncertScore"] - dat_model["uncertScore"].mean()
) / dat_model["uncertScore"].std()

dat_model.columns.tolist()
```

[9]:
```
['groupID',
 'participantID',
 'cf',
 'pp',
 'uncertScore',
 'realGamePeaks_per_m',
 'realTutoPeaks_per_m',
 'realQuestionsPeaks_per_m',
 'realPitchPeaks_per_m',
 'questionPath',
 'pitchRole',
 'scr_group_mean',
 'scr_within',
 'scr_between',
 'tuto_group_mean',
 'tuto_within',
 'tuto_between',
 'questions_group_mean',
 'questions_within',
 'questions_between',
 'pitch_group_mean',
 'pitch_within',
 'pitch_between',
 'pp_group_mean',
```

```
 'pp_within',
 'pp_between',
 'realGamePeaks_per_m_z',
 'scr_within_z',
 'scr_between_z',
 'tuto_within_z',
 'tuto_between_z',
 'questions_within_z',
 'questions_between_z',
 'pitch_within_z',
 'pitch_between_z',
 'pp_z',
 'pp_within_z',
 'pp_between_z',
 'uncertScore_z']
```

Pitch data cleaning

[10]:
```
dat_model['pitchRole'] = pd.to_numeric(dat_model['pitchRole'], errors='coerce')
dat_model_pitch_role = dat_model.dropna(subset=['groupID',
 'participantID',
 'cf',
 'pp',
 'uncertScore',
 'realGamePeaks_per_m',
 'realTutoPeaks_per_m',
 'realQuestionsPeaks_per_m',
 'realPitchPeaks_per_m',
 'questionPath',
 'pitchRole',
 'scr_group_mean',
 'scr_within',
 'scr_between',
 'tuto_group_mean',
 'tuto_within',
 'tuto_between',
 'questions_group_mean',
 'questions_within',
 'questions_between',
 'pitch_group_mean',
 'pitch_within',
 'pitch_between',
 'pp_group_mean',
 'pp_within',
 'pp_between',
 'realGamePeaks_per_m_z',
 'scr_within_z',
```

```
 'scr_between_z',
 'tuto_within_z',
 'tuto_between_z',
 'questions_within_z',
 'questions_between_z',
 'pitch_within_z',
 'pitch_between_z',
 'pp_z',
 'pp_within_z',
 'pp_between_z',
 'uncertScore_z']).copy()

dat_model_pitch_role = dat_model_pitch_role[dat_model_pitch_role['pitchRole'].
 ↪isin([0, 1])].copy()

# Finalize data type
dat_model_pitch_role["pitchRole"] = pd.
 ↪Categorical(dat_model_pitch_role["pitchRole"], categories=[0, 1])
```

1.2 Exploratory Data Analysis

[55]:
```
cols = ["cf", "pp", "uncertScore", "realGamePeaks_per_m",␣
 ↪"realTutoPeaks_per_m", "realQuestionsPeaks_per_m", "realPitchPeaks_per_m"]

# Labels to display
labels = {
    "cf": "Retrospective confidence",
    "pp": "Perceived individual performance",
    "pp_within": "Within-team PIP SCR frequency",
    "pp_between": "Between-team PIP SCR frequency",
    "uncertScore": "Team gameplay score",
    "realGamePeaks_per_m": "The whole GLS SCR frequency",
    "scr_within": "Within-team the whole GLS SCR frequency",
    "scr_between": "Between-team the whole GLS SCR frequency",
    "realTutoPeaks_per_m": "Tutorial SCR frequency",
    "tuto_within": "Within-team tutorial SCR frequency",
    "tuto_between": "Between-team tutorial SCR frequency",
    "realQuestionsPeaks_per_m": "Gameplay SCR frequency",
    "questions_within": "Within-team gameplay SCR frequency",
    "questions_between": "Between-team gameplay SCR frequency",
    "realPitchPeaks_per_m": "Pitch SCR frequency",
    "pitch_within": "Within-team pitch SCR frequency",
    "pitch_between": "Between-team pitch SCR frequency"
}

# Basic summary
print("Missing values:")
```

```
print(dat_model[cols].isna().sum(), "\n")

print("Descriptive statistics:")
display(dat_model[cols].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols), figsize=(20, 8))

for i, col in enumerate(cols):
    axes[0, i].hist(dat_model[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(dat_model[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()

# Correlation matrix
corr = dat_model[cols].corr()

fig, ax = plt.subplots(figsize=(8, 6))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)
ax.set_title("General Correlation Matrix")

# Add correlation scores in each cell
for i in range(len(cols)):
    for j in range(len(cols)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",␣
 ↪color=text_color)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```

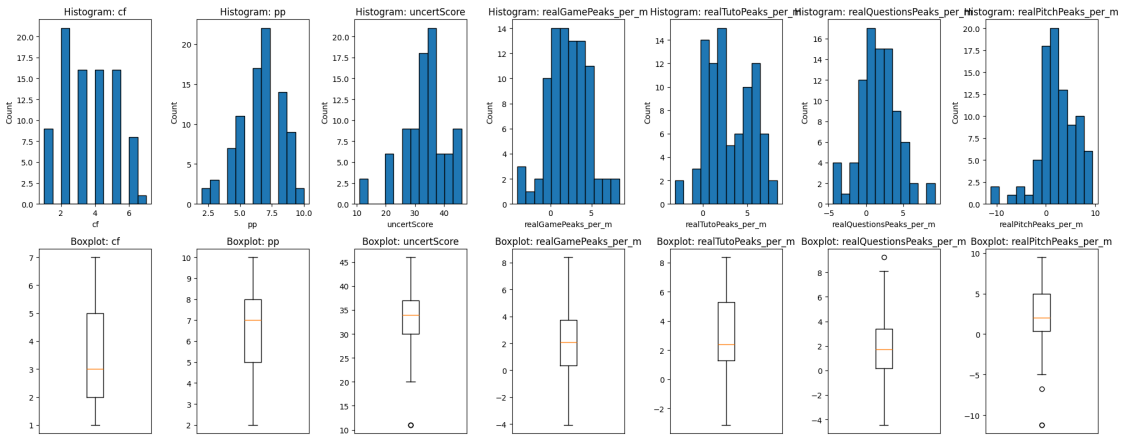
```
Missing values:
```

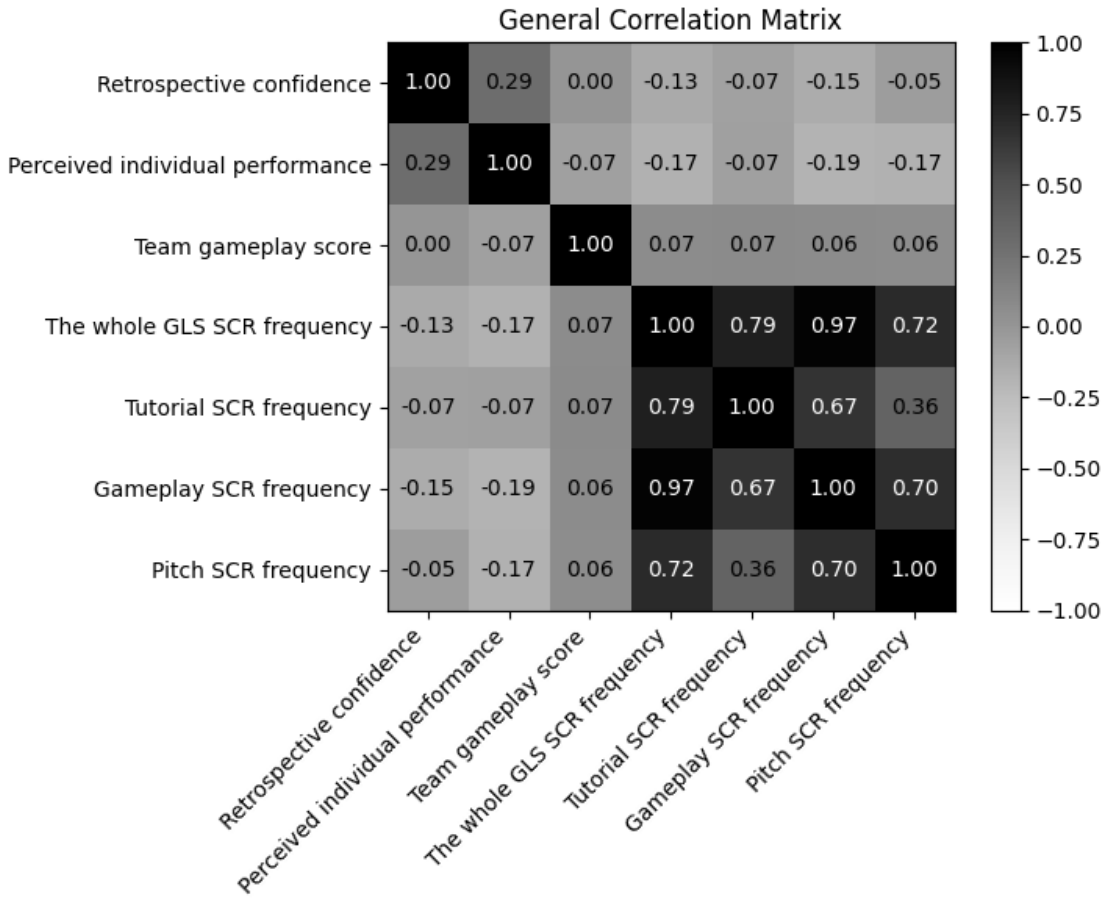
```
cf                          0
pp                          0
uncertScore                 0
realGamePeaks_per_m         0
realTutoPeaks_per_m         0
realQuestionsPeaks_per_m    0
realPitchPeaks_per_m        0
dtype: int64

Descriptive statistics:

                          count       mean       std        min        25%  \
cf                         87.0   3.425287  1.559906   1.000000   2.000000
pp                         87.0   6.494253  1.764466   2.000000   5.000000
uncertScore                87.0  33.482759  7.603338  11.000000  30.000000
realGamePeaks_per_m        87.0   2.114204  2.406057  -4.070671   0.358493
realTutoPeaks_per_m        87.0   2.989865  2.548684  -3.124891   1.270447
realQuestionsPeaks_per_m   87.0   1.800861  2.490374  -4.446577   0.189259
realPitchPeaks_per_m       87.0   2.272550  3.905028 -11.221374   0.346963

                            50%        75%        max
cf                     3.000000   5.000000   7.000000
pp                     7.000000   8.000000  10.000000
uncertScore           34.000000  37.000000  46.000000
realGamePeaks_per_m    2.079011   3.722039   8.403172
realTutoPeaks_per_m    2.378785   5.282077   8.361345
realQuestionsPeaks_per_m  1.723939   3.370871   9.234457
realPitchPeaks_per_m   2.019637   4.973874   9.502378
```





[48]:
```
eda_df = dat_model.copy()

eda_df[["pitch_within", "pitch_between"]] = np.nan
eda_df.loc[dat_model_pitch_role.index, ["pitch_within", "pitch_between"]] =␣
 ↪dat_model_pitch_role[["pitch_within", "pitch_between"]].to_numpy()

cols_within_between = ["cf", "pp", "uncertScore", "scr_within", "scr_between",␣
 ↪"tuto_within", "tuto_between","questions_within", "questions_between",␣
 ↪"pitch_within", "pitch_between"]
cols_within = ["cf", "pp", "uncertScore", "scr_within", "tuto_within",␣
 ↪"questions_within", "pitch_within"]
cols_between = ["cf", "pp", "uncertScore", "scr_between",  "tuto_between",␣
 ↪"questions_between", "pitch_between"]
```

Within-team Exploratory Data Analysis

[49]:
```
# Basic summary
print("Missing values:")
```

```
print(eda_df[cols_within].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_within].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_within), figsize=(20, 8))

for i, col in enumerate(cols_within):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

plt.tight_layout()
plt.show()
```

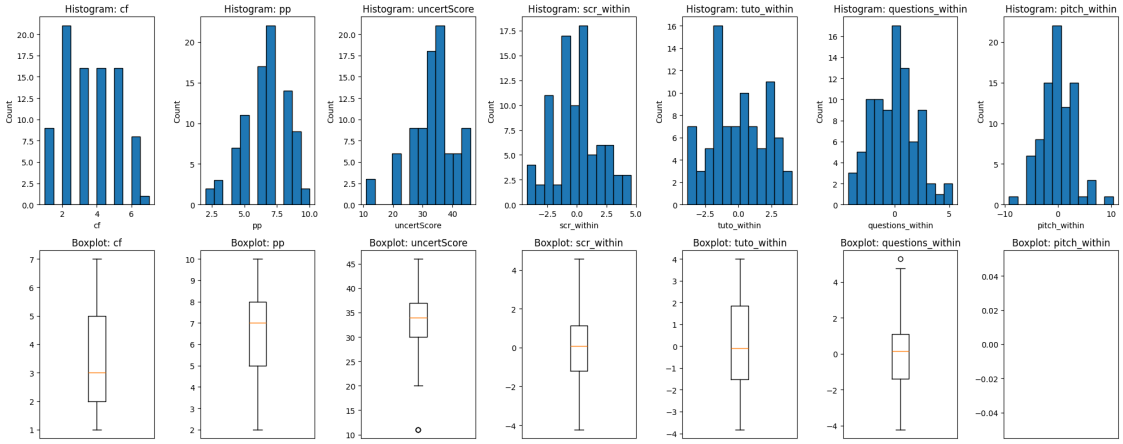
```
Missing values:
cf                  0
pp                  0
uncertScore         0
scr_within          0
tuto_within         0
questions_within    0
pitch_within        3
dtype: int64

Descriptive statistics:
```

	count	mean	std	min	25%	\
cf	87.0	3.425287e+00	1.559906	1.000000	2.000000	
pp	87.0	6.494253e+00	1.764466	2.000000	5.000000	
uncertScore	87.0	3.348276e+01	7.603338	11.000000	30.000000	
scr_within	87.0	4.083579e-17	1.952139	-4.228912	-1.186685	
tuto_within	87.0	1.020895e-17	2.049427	-3.827188	-1.513568	
questions_within	87.0	-1.786566e-17	1.992601	-4.223753	-1.400472	
pitch_within	84.0	-1.057355e-17	3.098589	-9.181356	-1.706938	

	50%	75%	max
cf	3.000000	5.000000	7.000000
pp	7.000000	8.000000	10.000000
uncertScore	34.000000	37.000000	46.000000
scr_within	0.079387	1.136715	4.572771
tuto_within	-0.096863	1.860652	4.003711

```
questions_within    0.153288    1.088501    5.281570
pitch_within       -0.176149    2.024036   10.406666
```



[56]:
```
# Correlation matrix
corr = eda_df[cols_within].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

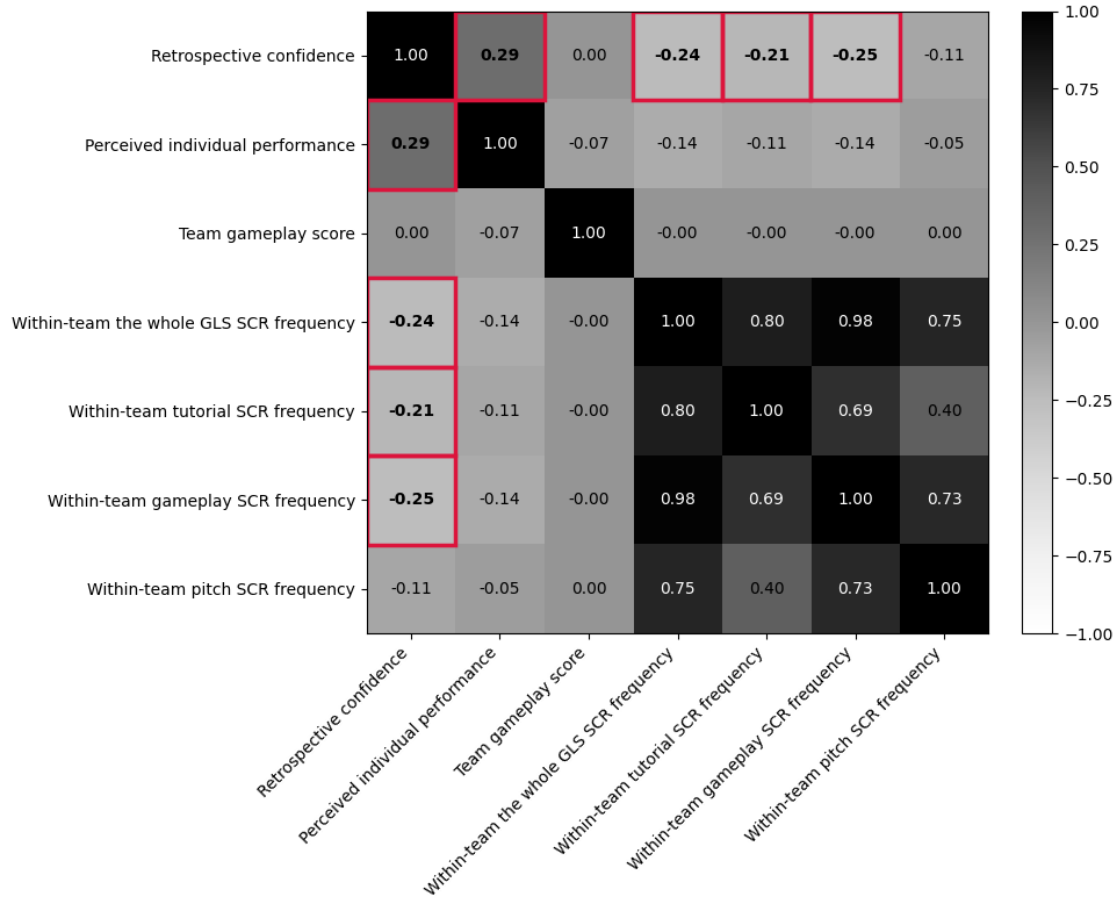
display_labels = [labels[col] for col in cols_within]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_within)):
    for j in range(len(cols_within)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within[i], cols_within[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)
```

```
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



Between-team Exploratory Data Analysis

[57]:
```
# Basic summary
print("Missing values:")
print(eda_df[cols_between].isna().sum(), "\n")

print("Descriptive statistics:")
display(eda_df[cols_between].describe().T)

# Distribution plots and boxplots
fig, axes = plt.subplots(2, len(cols_between), figsize=(20, 8))

for i, col in enumerate(cols_between):
    axes[0, i].hist(eda_df[col], bins=12, edgecolor="black")
```

```
    axes[0, i].set_title(f"Histogram: {col}")
    axes[0, i].set_xlabel(col)
    axes[0, i].set_ylabel("Count")

    axes[1, i].boxplot(eda_df[col], vert=True)
    axes[1, i].set_title(f"Boxplot: {col}")
    axes[1, i].set_xticks([])

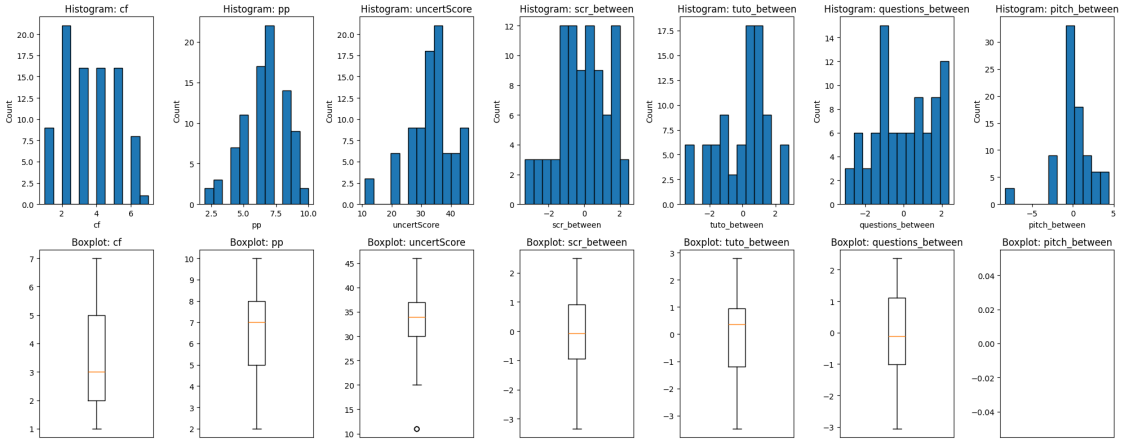
plt.tight_layout()
plt.show()
```

```
Missing values:
cf                   0
pp                   0
uncertScore          0
scr_between          0
tuto_between         0
questions_between    0
pitch_between        3
dtype: int64

Descriptive statistics:

                    count          mean       std        min        25%  \
cf                   87.0  3.425287e+00  1.559906   1.000000   2.000000
pp                   87.0  6.494253e+00  1.764466   2.000000   5.000000
uncertScore          87.0  3.348276e+01  7.603338  11.000000  30.000000
scr_between          87.0  2.756416e-16  1.406507  -3.337750  -0.941966
tuto_between         87.0 -8.473426e-16  1.515138  -3.478591  -1.187725
questions_between    87.0  8.167158e-17  1.493821  -3.056665  -1.009198
pitch_between        84.0  1.269501e-01  2.385979  -8.290591  -0.652798

                         50%        75%        max
cf                  3.000000   5.000000   7.000000
pp                  7.000000   8.000000  10.000000
uncertScore        34.000000  37.000000  46.000000
scr_between        -0.065864   0.914374   2.494181
tuto_between        0.369232   0.947484   2.792636
questions_between  -0.114830   1.106064   2.363263
pitch_between       0.139819   1.189454   4.415574
```



[58]:
```
# Correlation matrix
corr = eda_df[cols_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

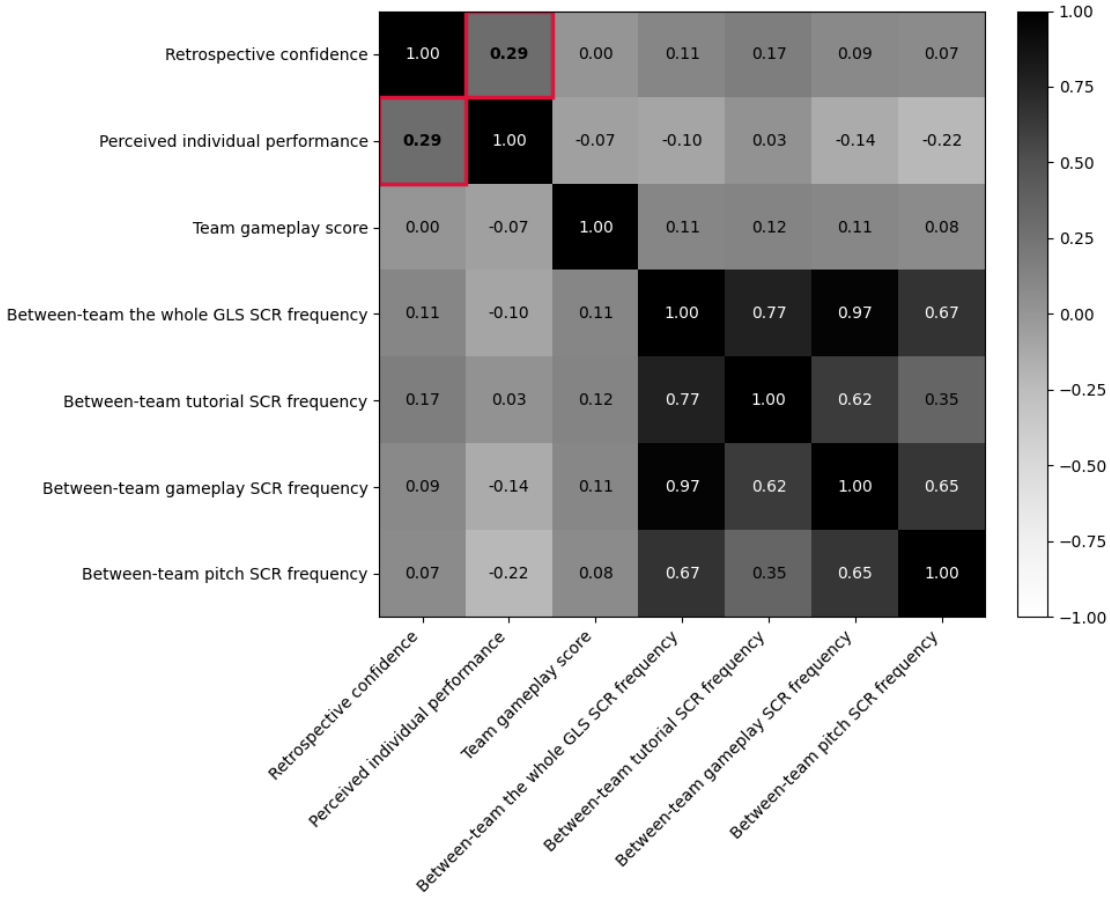
display_labels = [labels[col] for col in cols_between]

ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)
#ax.set_title("Detailed Correlation Matrix")

threshold = 0.20
focus_var = "cf"

for i in range(len(cols_between)):
    for j in range(len(cols_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_between[i], cols_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)
```

```
plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



[59]:
```
# Correlation matrix
corr = eda_df[cols_within_between].corr()

fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)

display_labels = [labels[col] for col in cols_within_between]

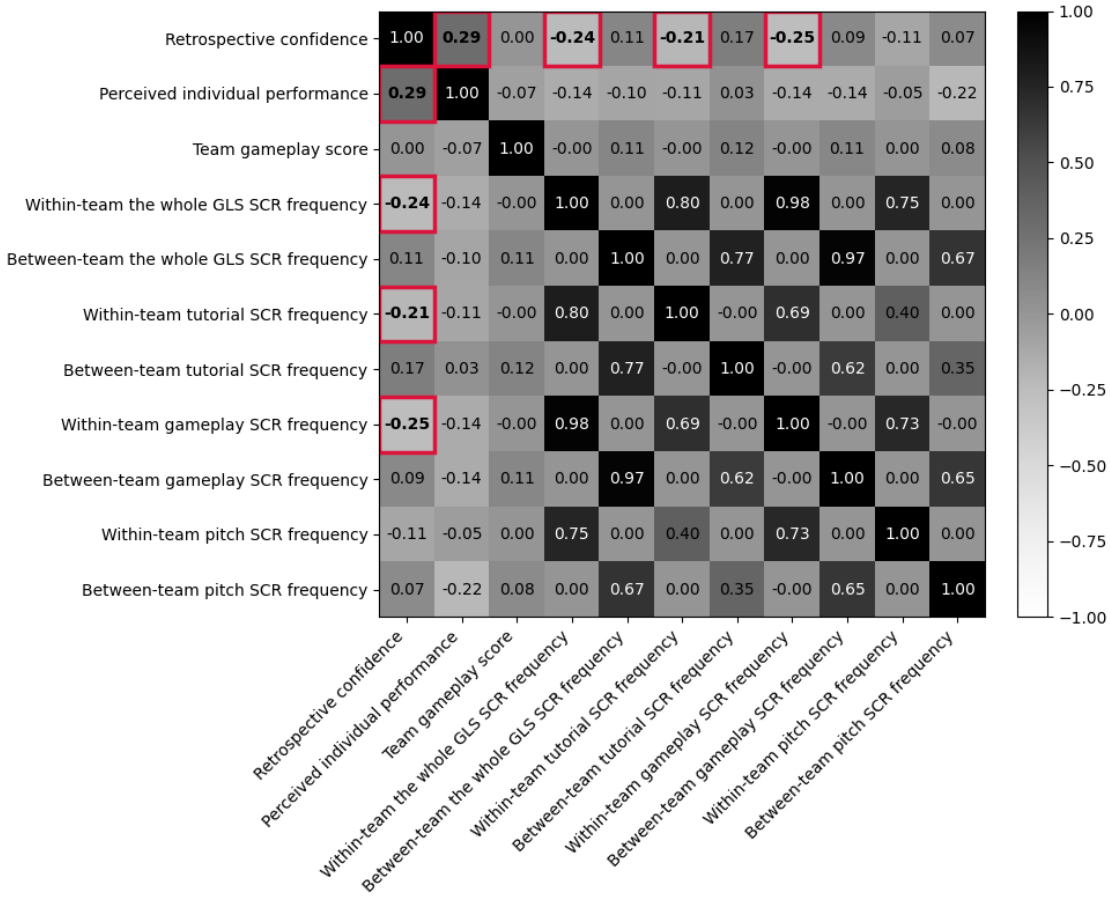
ax.set_xticks(range(len(display_labels)))
ax.set_yticks(range(len(display_labels)))
ax.set_xticklabels(display_labels, rotation=45, ha="right")
ax.set_yticklabels(display_labels)

threshold = 0.20
```

```
focus_var = "cf"

for i in range(len(cols_within_between)):
    for j in range(len(cols_within_between)):
        value = corr.iloc[i, j]
        text_color = "white" if abs(value) > 0.5 else "black"
        involves_focus = focus_var in (cols_within_between[i],␣
 ↪cols_within_between[j])
        is_strong = abs(value) >= threshold and i != j and involves_focus
        ax.text(j, i, f"{value:.2f}", ha="center", va="center",
                color=text_color, fontweight="bold" if is_strong else "normal")
        if is_strong:
            rect = patches.Rectangle((j - 0.5, i - 0.5), 1, 1,
                                     linewidth=2.5, edgecolor="crimson",␣
 ↪facecolor="none")
            ax.add_patch(rect)

plt.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
plt.tight_layout()
plt.show()
```



1.2.1 Model 1 - Game play SCR: CF ~ realGamePeaks_per_m_z + (1|groupID)

1. Statsmodels (LME4)

[18]:
```
model1_sm = sm.MixedLM.from_formula(
    "cf ~ realGamePeaks_per_m_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_sm = model1_sm.fit(reml=True, method="lbfgs")
print("Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)")
print(result1_sm.summary())
```

```
Model 1 sm: CF ~ realGamePeaks_per_m_z + (1|groupID)
             Mixed Linear Model Regression Results
================================================================
Model:               MixedLM    Dependent Variable:    cf
No. Observations:    87         Method:                REML
No. Groups:          29         Scale:                 1.8671
Min. group size:     3          Log-Likelihood:        -160.6816
Max. group size:     3          Converged:             Yes
Mean group size:     3.0
----------------------------------------------------------------
                      Coef.  Std.Err.   z    P>|z| [0.025 0.975]
----------------------------------------------------------------
Intercept              3.425    0.203 16.854 0.000  3.027  3.824
realGamePeaks_per_m_z -0.298    0.167 -1.789 0.074 -0.625  0.029
groupID Var            0.575    0.295
================================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[19]:
```
model1_bmb = bmb.Model(
    "cf ~ realGamePeaks_per_m_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1 = model1_bmb.fit(
```

```
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)")
az.summary(idata_model1, var_names=["realGamePeaks_per_m_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, realGamePeaks_per_m_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 1 bmb: CF ~ realGamePeaks_per_m_z + (1|groupID)
```

[19]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
realGamePeaks_per_m_z	-0.293	0.174	-0.606	0.042	0.002	0.002	
1|groupID_sigma	0.727	0.277	0.149	1.217	0.007	0.004	
sigma	1.405	0.134	1.172	1.670	0.003	0.002	

	ess_bulk	ess_tail	r_hat
realGamePeaks_per_m_z	5495.0	4901.0	1.0
1|groupID_sigma	1695.0	1804.0	1.0
sigma	2684.0	3775.0	1.0

Within/Between Model 1: CF ~ scr_within_z + scr_between_z + (1|groupID)

1. Statsmodels (LME4)

[20]:
```
model1_within_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_sm = model1_within_sm.fit(reml=True, method="lbfgs")
print("Model 1 within sm: CF ~ scr_within_z + (1|groupID)")
print(result1_within_sm.summary())
```

```
Model 1 within sm: CF ~ scr_within_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              1.8480
Min. group size:  3       Log-Likelihood:     -159.2149
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
             Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.194 17.632 0.000  3.045  3.806
scr_within_z -0.374    0.147 -2.548 0.011 -0.661 -0.086
groupID Var   0.478    0.263
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[21]:
```
model1_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_between_sm = model1_between_sm.fit(reml=True, method="lbfgs")
print("Model 1 between sm: CF ~ scr_between_z + (1|groupID)")
print(result1_between_sm.summary())
```

```
Model 1 between sm: CF ~ scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -161.6428
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.195 17.573 0.000  3.043  3.807
scr_between_z 0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var   0.427    0.255
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[22]:
```
model1_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ scr_within_z + scr_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result1_within_between_sm = model1_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 1 within-between sm: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
print(result1_within_between_sm.summary())
```

```
Model 1 within-between sm: CF ~ scr_within_z + scr_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8480
Min. group size:   3       Log-Likelihood:     -159.5196
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef.  Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      3.425    0.195 17.573 0.000  3.043  3.807
scr_within_z  -0.374    0.147 -2.548 0.011 -0.661 -0.086
scr_between_z  0.177    0.196  0.901 0.368 -0.208  0.561
groupID Var    0.486    0.268
=========================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[23]:
```
model1_within_bmb  = bmb.Model(
    "cf ~ scr_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
```

```
)

idata_model1_within_bmb = model1_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 1 within bmb: CF ~ scr_within_z + (1|groupID)")
az.summary(idata_model1_within_bmb, var_names=["scr_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.
There were 64 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 1 within bmb: CF ~ scr_within_z + (1|groupID)
```

[23]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
scr_within_z	-0.371	0.153	-0.660	-0.085	0.002	0.002	4811.0	
1|groupID_sigma	0.644	0.272	0.059	1.098	0.007	0.004	1451.0	
sigma	1.399	0.132	1.175	1.658	0.003	0.002	2406.0	

	ess_tail	r_hat
scr_within_z	4477.0	1.0
1|groupID_sigma	1908.0	1.0
sigma	3769.0	1.0

[24]:
```
model1_between_bmb  = bmb.Model(
    "cf ~ scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_between_bmb = model1_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
```

```
    random_seed=123,
    target_accept=0.99
)

print("Model 1 between bmb: CF ~ scr_between_z + (1|groupID)")
az.summary(idata_model1_between_bmb, var_names=["scr_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 1 between bmb: CF ~ scr_between_z + (1|groupID)
```

[24]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
scr_between_z	0.173	0.201	-0.207	0.545	0.003	0.002	6227.0	5718.0	1.0
1|groupID_sigma	0.611	0.279	0.012	1.041	0.007	0.003	1454.0	1929.0	1.0
sigma	1.460	0.133	1.217	1.713	0.002	0.001	3505.0	4732.0	1.0

[25]:
```
model1_within_between_bmb  = bmb.Model(
    "cf ~ scr_within_z + scr_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model1_within_between_bmb = model1_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model1_within_between_bmb, var_names=["scr_within_z",␣
 ↪"scr_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, scr_within_z, scr_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 1 within-between bmb: CF ~ scr_within_z + scr_between_z + (1|groupID)
```

[25]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
scr_within_z	-0.373	0.149	-0.647	-0.090	0.001	0.002	11983.0	
scr_between_z	0.179	0.201	-0.184	0.559	0.003	0.002	5997.0	
1|groupID_sigma	0.663	0.275	0.082	1.145	0.007	0.004	1406.0	
sigma	1.397	0.131	1.159	1.640	0.002	0.001	2930.0	

	ess_tail	r_hat
scr_within_z	5488.0	1.0
scr_between_z	5593.0	1.0
1|groupID_sigma	1239.0	1.0
sigma	5151.0	1.0

1.2.2 Model 2 - Event Phase: CF ~ event_phase_within/between_z + (1|groupID)

a) Tutorial

1. Statsmodels (LME4)

[26]:
```
model2_tuto_within_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_sm = model2_tuto_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)")
print(result2_tuto_within_sm.summary())
```

```
Model 2 tuto within sm: CF ~ tuto_within_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              1.8934
Min. group size:   3       Log-Likelihood:     -159.9074
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
```

```
------------------------------------------------------
              Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------
Intercept      3.425    0.194 17.632 0.000  3.045  3.806
tuto_within_z -0.331    0.148 -2.229 0.026 -0.622 -0.040
groupID Var    0.463    0.260
======================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[27]:
```
model2_tuto_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_between_sm = model2_tuto_between_sm.fit(reml=True, method="lbfgs")
print("Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)")
print(result2_tuto_between_sm.summary())
```

```
Model 2 tuto between sm: CF ~ tuto_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.0230
Min. group size:   3       Log-Likelihood:     -161.1722
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
               Coef. Std.Err.    z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept      3.425    0.192 17.882 0.000  3.050  3.801
tuto_between_z 0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var    0.390    0.246
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[28]:
```
model2_tuto_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ tuto_within_z + tuto_between_z",
```

```
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_tuto_within_between_sm = model2_tuto_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
print(result2_tuto_within_between_sm.summary())
```

```
Model 2 tuto within-between sm: CF ~ tuto_within_z + tuto_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   87      Method:             REML
No. Groups:         29      Scale:              1.8934
Min. group size:    3       Log-Likelihood:     -159.7416
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.425    0.192 17.882 0.000  3.050  3.801
tuto_within_z  -0.331    0.148 -2.229 0.026 -0.622 -0.040
tuto_between_z  0.258    0.193  1.341 0.180 -0.119  0.636
groupID Var     0.433    0.255
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[29]:
```
model2_tuto_within_bmb  = bmb.Model(
    "cf ~ tuto_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_bmb = model2_tuto_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
```

```
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)")
az.summary(idata_model2_tuto_within_bmb, var_names=["tuto_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 2 tuto within bmb: CF ~ tuto_within_z + (1|groupID)
```

[29]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
tuto_within_z	-0.331	0.153	-0.614	-0.038	0.001	0.002	10654.0	
1|groupID_sigma	0.631	0.277	0.049	1.095	0.008	0.004	1306.0	
sigma	1.419	0.138	1.176	1.681	0.003	0.002	2809.0	

	ess_tail	r_hat
tuto_within_z	6259.0	1.00
1|groupID_sigma	1451.0	1.01
sigma	4823.0	1.00

[30]:
```
model2_tuto_between_bmb  = bmb.Model(
    "cf ~ tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_between_bmb = model2_tuto_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)")
az.summary(idata_model2_tuto_between_bmb, var_names=["tuto_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.
```

```
Model 2 tuto between bmb: CF ~ tuto_between_z + (1|groupID)
```

[30]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
tuto_between_z	0.257	0.196	-0.102	0.642	0.003	0.003	5969.0	5659.0	1.0
1|groupID_sigma	0.582	0.283	0.002	1.017	0.007	0.003	1406.0	1613.0	1.0
sigma	1.458	0.134	1.213	1.710	0.002	0.001	3579.0	5319.0	1.0

[31]:
```
model2_tuto_within_between_bmb  = bmb.Model(
    "cf ~ tuto_within_z + tuto_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_tuto_within_between_bmb = model2_tuto_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model2_tuto_within_between_bmb, var_names=["tuto_within_z",␣
 ↪"tuto_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, tuto_within_z, tuto_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 2 tuto within-between bmb: CF ~ tuto_within_z + tuto_between_z +
(1|groupID)
```

[31]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	\
tuto_within_z	-0.331	0.154	-0.618	-0.042	0.001	0.002	11871.0	
tuto_between_z	0.259	0.197	-0.116	0.630	0.003	0.002	6119.0	
1|groupID_sigma	0.615	0.276	0.018	1.046	0.008	0.004	1324.0	
sigma	1.414	0.133	1.170	1.658	0.002	0.001	3106.0	

	ess_tail	r_hat
tuto_within_z	5929.0	1.0
tuto_between_z	5801.0	1.0
1|groupID_sigma	1400.0	1.0
sigma	5239.0	1.0

b) Questions

1. Statsmodels (LME4)

[32]:
```
model2_questions_within_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_sm = model2_questions_within_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)")
print(result2_questions_within_sm.summary())
```

```
Model 2 questions within sm: CF ~ questions_within_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                1.8232
Min. group size:    3         Log-Likelihood:       -158.8308
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                   Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.194 17.632 0.000  3.045  3.806
questions_within_z -0.395    0.146 -2.712 0.007 -0.680 -0.109
groupID Var         0.487    0.265
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[33]:
```
model2_questions_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_between_sm = model2_questions_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)")
print(result2_questions_between_sm.summary())
```

```
Model 2 questions between sm: CF ~ questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:              MixedLM   Dependent Variable:   cf
No. Observations:   87        Method:               REML
No. Groups:         29        Scale:                2.0230
Min. group size:    3         Log-Likelihood:       -161.7841
Max. group size:    3         Converged:            Yes
Mean group size:    3.0
------------------------------------------------------------
                    Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept           3.425    0.196 17.481 0.000  3.041  3.809
questions_between_z 0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var         0.439    0.258
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[34]:
```
model2_questions_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ questions_within_z + questions_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result2_questions_within_between_sm = model2_questions_within_between_sm.
 ↪fit(reml=True, method="lbfgs")
```

```
print("Model 2 questions within-between sm: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
print(result2_questions_within_between_sm.summary())
```

```
Model 2 questions within-between sm: CF ~ questions_within_z +
questions_between_z + (1|groupID)
            Mixed Linear Model Regression Results
============================================================
Model:               MixedLM   Dependent Variable:   cf
No. Observations:    87        Method:               REML
No. Groups:          29        Scale:                1.8232
Min. group size:     3         Log-Likelihood:       -159.2768
Max. group size:     3         Converged:            Yes
Mean group size:     3.0
------------------------------------------------------------
                    Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept            3.425    0.196 17.481 0.000  3.041  3.809
questions_within_z  -0.395    0.146 -2.712 0.007 -0.680 -0.110
questions_between_z  0.142    0.197  0.723 0.470 -0.244  0.529
groupID Var          0.506    0.272
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[35]:
```
model2_questions_within_bmb  = bmb.Model(
    "cf ~ questions_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_bmb = model2_questions_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)")
az.summary(idata_model2_questions_within_bmb, var_names=["questions_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There were 52 divergences after tuning. Increase `target_accept` or
reparameterize.
```

```
Model 2 questions within bmb: CF ~ questions_within_z + (1|groupID)
```

[35]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	\
questions_within_z	-0.396	0.152	-0.697	-0.123	0.002	0.003	
1|groupID_sigma	0.659	0.267	0.119	1.144	0.008	0.004	
sigma	1.389	0.131	1.147	1.625	0.003	0.002	

	ess_bulk	ess_tail	r_hat
questions_within_z	4089.0	2600.0	1.0
1|groupID_sigma	1099.0	2147.0	1.0
sigma	1587.0	1906.0	1.0

[36]:
```
model2_questions_between_bmb  = bmb.Model(
    "cf ~ questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_between_bmb = model2_questions_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_between_bmb,␣
 ↪var_names=["questions_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.

Model 2 questions between bmb: CF ~ questions_between_z + (1|groupID)
```

[36]:
```
                       mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
questions_between_z   0.141  0.200  -0.237    0.513      0.003    0.002
1|groupID_sigma       0.619  0.283   0.000    1.044      0.007    0.004
sigma                 1.458  0.134   1.213    1.709      0.002    0.002

                      ess_bulk  ess_tail  r_hat
questions_between_z     5297.0    5488.0    1.0
1|groupID_sigma         1511.0    1691.0    1.0
sigma                   3060.0    4972.0    1.0
```

[37]:
```
model2_questions_within_between_bmb  = bmb.Model(
    "cf ~ questions_within_z + questions_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model2_questions_within_between_bmb = model2_questions_within_between_bmb.
 ↪fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 questions within-between bmb: CF ~ questions_within_z +␣
 ↪questions_between_z + (1|groupID)")
az.summary(idata_model2_questions_within_between_bmb,␣
 ↪var_names=["questions_within_z", "questions_between_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questions_within_z, questions_between_z,
1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 4 seconds.

Model 2 questions within-between bmb: CF ~ questions_within_z +
questions_between_z + (1|groupID)
```

[37]:
```
                       mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
questions_within_z   -0.395  0.150  -0.677   -0.105      0.001    0.002
questions_between_z   0.141  0.203  -0.264    0.501      0.003    0.002
1|groupID_sigma       0.675  0.272   0.079    1.135      0.007    0.004
sigma                 1.387  0.131   1.142    1.627      0.002    0.001

                     ess_bulk  ess_tail  r_hat
questions_within_z    12271.0    6170.0    1.0
questions_between_z    4990.0    4878.0    1.0
1|groupID_sigma        1467.0    1504.0    1.0
sigma                  3354.0    4823.0    1.0
```

c) Pitch

1. Statsmodels (LME4)

[38]:
```
model2_pitch_within_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_sm = model2_pitch_within_sm.fit(reml=True, method="lbfgs")
print("Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)")
print(result2_pitch_within_sm.summary())
```

```
Model 2 pitch within sm: CF ~ pitch_within_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9327
Min. group size:    3       Log-Likelihood:     -155.4275
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.417    0.201 16.987 0.000  3.022  3.811
pitch_within_z -0.166    0.150 -1.107 0.268 -0.461  0.128
groupID Var     0.488    0.271
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[39]:
```
model2_pitch_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_between_sm = model2_pitch_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)")
print(result2_pitch_between_sm.summary())
```

```
Model 2 pitch between sm: CF ~ pitch_between_z + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.9405
Min. group size:    3       Log-Likelihood:     -155.5600
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.411    0.204 16.713 0.000  3.011  3.811
pitch_between_z 0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var     0.516    0.280
============================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[40]:
```
model2_pitch_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ pitch_within_z + pitch_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
result2_pitch_within_between_sm = model2_pitch_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +␣
 ↪(1|groupID)")
print(result2_pitch_within_between_sm.summary())
```

```
Model 2 pitch within-between sm: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
          Mixed Linear Model Regression Results
```

```
============================================================
Model:              MixedLM  Dependent Variable:  cf
No. Observations:   84       Method:              REML
No. Groups:         28       Scale:               1.9327
Min. group size:    3        Log-Likelihood:      -155.9245
Max. group size:    3        Converged:           Yes
Mean group size:    3.0
------------------------------------------------------------
                Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept        3.411    0.204 16.713 0.000  3.011  3.811
pitch_within_z  -0.166    0.150 -1.107 0.268 -0.461  0.128
pitch_between_z  0.115    0.210  0.550 0.582 -0.296  0.526
groupID Var      0.518    0.281
============================================================
```

```
/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[41]:
```
model2_pitch_within_bmb  = bmb.Model(
    "cf ~ pitch_within_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_bmb = model2_pitch_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)")
az.summary(idata_model2_pitch_within_bmb, var_names=["pitch_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
```

```
total) took 2 seconds.
There were 187 divergences after tuning. Increase `target_accept` or
reparameterize.

Model 2 pitch within bmb: CF ~ pitch_within_z + (1|groupID)
```

[41]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
pitch_within_z	-0.166	0.156	-0.457	0.137	0.002	0.002	5687.0	4425.0	1.0
1|groupID_sigma	0.655	0.279	0.071	1.128	0.008	0.003	1315.0	2591.0	1.0
sigma	1.425	0.144	1.169	1.700	0.004	0.003	1159.0	576.0	1.0

[42]:
```
model2_pitch_between_bmb  = bmb.Model(
    "cf ~ pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_between_bmb = model2_pitch_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)")
az.summary(idata_model2_pitch_between_bmb, var_names=["pitch_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_between_z, 1|groupID_sigma, 1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.

Model 2 pitch between bmb: CF ~ pitch_between_z + (1|groupID)
```

[42]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk
pitch_between_z	0.112	0.219	-0.290	0.529	0.003	0.002	5771.0
1|groupID_sigma	0.675	0.291	0.074	1.182	0.008	0.004	1434.0

```
sigma            1.435  0.138   1.184    1.690      0.003    0.002     2902.0

                 ess_tail  r_hat
pitch_between_z    6026.0    1.0
1|groupID_sigma    1321.0    1.0
sigma              4797.0    1.0
```

[43]:
```
model2_pitch_within_between_bmb  = bmb.Model(
    "cf ~ pitch_within_z + pitch_between_z + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_model2_pitch_within_between_bmb = model2_pitch_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z␣
 ↪+ (1|groupID)")
az.summary(idata_model2_pitch_within_between_bmb, var_names=["pitch_within_z",␣
 ↪"pitch_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitch_within_z, pitch_between_z, 1|groupID_sigma,
1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
There was 1 divergence after tuning. Increase `target_accept` or reparameterize.
```

```
Model 2 pitch within-between bmb: CF ~ pitch_within_z + pitch_between_z +
(1|groupID)
```

[43]:
```
                  mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  \
pitch_within_z  -0.165  0.155  -0.462    0.118      0.002    0.002    8711.0
pitch_between_z  0.116  0.220  -0.322    0.506      0.003    0.003    4343.0
1|groupID_sigma  0.676  0.289   0.000    1.108      0.008    0.004    1337.0
sigma            1.433  0.139   1.184    1.693      0.003    0.001    2777.0

                 ess_tail  r_hat
pitch_within_z     5469.0    1.0
```

```
pitch_between_z     3655.0    1.0
1|groupID_sigma     1748.0    1.0
sigma               4716.0    1.0
```

1.2.3 Model 3.0 - Performance perception: CF ~ pp_z + (1|groupID)

1. Statsmodels (LME4)

[44]:
```
model3_0_sm = sm.MixedLM.from_formula(
    "cf ~ pp_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_0_sm = model3_0_sm.fit(reml=True, method="lbfgs")
print("Model 3_0 sm: CF ~ pp_z + (1|groupID)")
print(result3_0_sm.summary())
```

```
Model 3_0 sm: CF ~ pp_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:             MixedLM Dependent Variable: cf
No. Observations:  87      Method:             REML
No. Groups:        29      Scale:              2.1344
Min. group size:   3       Log-Likelihood:     -159.5135
Max. group size:   3       Converged:          Yes
Mean group size:   3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.170 20.175 0.000  3.093  3.758
pp_z          0.421    0.178  2.369 0.018  0.073  0.769
groupID Var   0.124    0.202
=========================================================


/Users/elvisortegaec/Downloads/data_analysis/.venv/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[45]:
```
model3_0_bmb  = bmb.Model(
    "cf ~ pp_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)
```

```
idata_model3_0_bmb = model3_0_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3_0 bmb: CF ~ pp_z + (1|groupID)")
az.summary(idata_model3_0_bmb, var_names=["pp_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 3_0 bmb: CF ~ pp_z + (1|groupID)
```

[45]:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
pp_z	0.405	0.173	0.079	0.728	0.002	0.002	7537.0	6241.0	1.0
1|groupID_sigma	0.393	0.250	0.000	0.812	0.006	0.002	1757.0	2396.0	1.0
sigma	1.469	0.123	1.245	1.703	0.002	0.001	4978.0	5902.0	1.0

1.2.4 Model 3 - Performance Perception: CF ~ pp_within/between_z + (1|groupID)

1. Statsmodels (LME4)

[51]:
```
model3_pp_within_sm = sm.MixedLM.from_formula(
    "cf ~ pp_within_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_within_sm = model3_pp_within_sm.fit(reml=True, method="lbfgs")
print("Model 3 pp within sm: CF ~ pp_within_z + (1|groupID)")
print(result3_pp_within_sm.summary())
```

```
Model 3 pp within sm: CF ~ pp_within_z + (1|groupID)
         Mixed Linear Model Regression Results
```

```
============================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0545
Min. group size:  3       Log-Likelihood:     -162.2339
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept     3.425    0.194 17.632 0.000  3.045  3.806
pp_within_z   0.052    0.155  0.334 0.738 -0.251  0.355
groupID Var   0.410    0.249
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[52]:
```
model3_pp_between_sm = sm.MixedLM.from_formula(
    "cf ~ pp_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_between_sm = model3_pp_between_sm.fit(reml=True, method="lbfgs")
print("Model 3 pp between sm: CF ~ pp_between_z + (1|groupID)")
print(result3_pp_between_sm.summary())
```

```
Model 3 pp between sm: CF ~ pp_between_z + (1|groupID)
         Mixed Linear Model Regression Results
============================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -156.4649
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
------------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept     3.425    0.161 21.288 0.000  3.110  3.741
pp_between_z  0.601    0.162  3.717 0.000  0.284  0.919
groupID Var   0.076    0.174
============================================================
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

[53]:
```
model3_pp_within_between_sm = sm.MixedLM.from_formula(
    "cf ~ pp_within_z + pp_between_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result3_pp_within_between_sm = model3_pp_within_between_sm.fit(reml=True,␣
 ↪method="lbfgs")
print("Model 3 pp within-between sm: CF ~ pp_within_z + pp_between_z +␣
 ↪(1|groupID)")
print(result3_pp_within_between_sm.summary())
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

```
Model 3 pp within-between sm: CF ~ pp_within_z + pp_between_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0545
Min. group size:  3       Log-Likelihood:     -157.3608
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.161 21.288 0.000  3.110  3.741
pp_within_z   0.052    0.155  0.334 0.738 -0.251  0.355
pp_between_z  0.601    0.162  3.717 0.000  0.284  0.919
groupID Var   0.066    0.173
=========================================================
```

2. Bambi (Robustness testing)

[54]:
```
model3_pp_within_bmb  = bmb.Model(
    "cf ~ pp_within_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)
```

```
idata_model3_pp_within_bmb = model3_pp_within_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3 pp within bmb: CF ~ pp_within_z + (1|groupID)")
az.summary(idata_model3_pp_within_bmb, var_names=["pp_within_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_within_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 3 seconds.
```

```
Model 3 pp within bmb: CF ~ pp_within_z + (1|groupID)
```

[54]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_within_z	0.051	0.16	-0.2	0.31	10477	6148	1.00	
1|groupID_sigma	0.6	0.281	0.12	1	1597	2047	1.00	
sigma	1.47	0.134	1.3	1.7	3416	4511	1.00	

	mcse_mean	mcse_sd
pp_within_z	0.0016	0.0011
1|groupID_sigma	0.007	0.0047
sigma	0.0023	0.0017

[55]:
```
model3_pp_between_bmb  = bmb.Model(
    "cf ~ pp_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model3_pp_between_bmb = model3_pp_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)
```

```
print("Model 3 pp between bmb: CF ~ pp_between_z + (1|groupID)")
az.summary(idata_model3_pp_between_bmb, var_names=["pp_between_z",␣
 ↪"1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pp_between_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 3 pp between bmb: CF ~ pp_between_z + (1|groupID)
```

[55]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_between_z	0.599	0.174	0.33	0.88	6738	5835	1.00	
1|groupID_sigma	0.36	0.229	0.038	0.75	1761	2535	1.00	
sigma	1.424	0.119	1.2	1.6	5328	5161	1.00	

	mcse_mean	mcse_sd
pp_between_z	0.0021	0.0016
1|groupID_sigma	0.0051	0.0035
sigma	0.0016	0.0012

[56]:
```
model3_pp_within_between_bmb  = bmb.Model(
    "cf ~ pp_within_z + pp_between_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model3_pp_within_between_bmb = model3_pp_within_between_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 3 pp within-between bmb: CF ~ pp_within_z + pp_between_z +␣
 ↪(1|groupID)")
az.summary(idata_model3_pp_within_between_bmb, var_names=["pp_within_z",␣
 ↪"pp_between_z", "1|groupID_sigma", "sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
```

```
NUTS: [sigma, Intercept, pp_within_z, pp_between_z, 1|groupID_sigma,
1|groupID_offset]

Output()


Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.

Model 3 pp within-between bmb: CF ~ pp_within_z + pp_between_z + (1|groupID)
```

[56]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pp_within_z	0.053	0.155	-0.2	0.3	13155	6272	1.00	
pp_between_z	0.602	0.173	0.33	0.88	9215	6051	1.00	
1|groupID_sigma	0.346	0.227	0.034	0.74	2136	2724	1.00	
sigma	1.434	0.119	1.3	1.6	5596	5626	1.00	

	mcse_mean	mcse_sd
pp_within_z	0.0014	0.00095
pp_between_z	0.0018	0.0013
1|groupID_sigma	0.0047	0.0032
sigma	0.0016	0.0012

1.2.5 Model 4 - Uncertainty Score: CF ~ uncertScore + (1|groupID)

1. Statsmodels (LME4)

[57]:
```
model4_sm = sm.MixedLM.from_formula(
    "cf ~ uncertScore_z",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
result4_sm = model4_sm.fit(reml=True, method="lbfgs")
print("Model 4 sm: CF ~ uncertScore_z + (1|groupID)")
print(result4_sm.summary())
```

```
Model 4 sm: CF ~ uncertScore_z + (1|groupID)
         Mixed Linear Model Regression Results
=========================================================
Model:            MixedLM Dependent Variable: cf
No. Observations: 87      Method:             REML
No. Groups:       29      Scale:              2.0230
Min. group size:  3       Log-Likelihood:     -162.0425
Max. group size:  3       Converged:          Yes
Mean group size:  3.0
---------------------------------------------------------
              Coef. Std.Err.   z    P>|z| [0.025 0.975]
---------------------------------------------------------
Intercept     3.425    0.198 17.315 0.000  3.038  3.813
```

```
uncertScore_z 0.005    0.199  0.024 0.981 -0.385  0.395
groupID Var   0.461    0.263
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[58]:
```
model4_bmb  = bmb.Model(
    "cf ~ uncertScore_z + (1|groupID)",
    data=dat_model,
    family="gaussian"
)

idata_model4_bmb = model4_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model 4 bmb: CF ~ uncertScore_z + (1|groupID)")
az.summary(idata_model4_bmb, var_names=["uncertScore_z", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, uncertScore_z, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model 4 bmb: CF ~ uncertScore_z + (1|groupID)
```

[58]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
uncertScore_z	0.008	0.204	-0.31	0.34	5738	5586	1.00	
1|groupID_sigma	0.65	0.277	0.17	1.1	1533	1366	1.00	
sigma	1.455	0.133	1.3	1.7	3469	3945	1.00	

	mcse_mean	mcse_sd
uncertScore_z	0.0027	0.002

```
1|groupID_sigma     0.007  0.0048
sigma              0.0023  0.0017
```

1.2.6 Follow-up models

Model Follow-up 1 Questions Path: CF ~ questionPath + (1|groupID)

1. Statsmodels (LME4)

[59]:
```
modelFU1_sm = sm.MixedLM.from_formula(
    "cf ~ questionPath",
    groups="groupID",
    re_formula="1",
    data=dat_model
)
resultFU1_sm = modelFU1_sm.fit(reml=True, method="lbfgs")
print("Model FU1 sm: CF ~ questionPath + (1|groupID)")
print(resultFU1_sm.summary())
```

```
Model FU1 sm: CF ~ questionPath + (1|groupID)
           Mixed Linear Model Regression Results
============================================================
Model:               MixedLM  Dependent Variable:  cf
No. Observations:    87       Method:              REML
No. Groups:          29       Scale:               2.0230
Min. group size:     3        Log-Likelihood:      -161.1184
Max. group size:     3        Converged:           Yes
Mean group size:     3.0
------------------------------------------------------------
                  Coef.  Std.Err.   z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept          3.500    0.237 14.779 0.000  3.036  3.964
questionPath[T.1] -0.241    0.425 -0.566 0.571 -1.074  0.592
groupID Var        0.447    0.260
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[60]:
```
modelFU1_bmb  = bmb.Model(
    "cf ~ questionPath + (1|groupID)",
    data=dat_model,
    family="gaussian"
)
```

```
idata_modelFU1_bmb = modelFU1_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123,
    target_accept=0.99
)

print("Model FU1 bmb: CF ~ questionPath + (1|groupID)")
az.summary(idata_modelFU1_bmb, var_names=["questionPath", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, questionPath, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 2 seconds.
```

```
Model FU1 bmb: CF ~ questionPath + (1|groupID)
```

[60]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
questionPath[1]	-0.25	0.439	-0.95	0.45	6070	5460	1.00	
1|groupID_sigma	0.63	0.29	0.13	1.1	1094	967	1.00	
sigma	1.458	0.134	1.3	1.7	3191	4209	1.00	

	mcse_mean	mcse_sd
questionPath[1]	0.0056	0.0041
1|groupID_sigma	0.0086	0.0058
sigma	0.0024	0.0017

Model Follow-up 2 PitchRole: CF ~ pitchRole + (1|groupID)

1. Statsmodels (LME4)

[61]:
```
modelFU2_sm = sm.MixedLM.from_formula(
    "cf ~ pitchRole",
    groups="groupID",
    re_formula="1",
    data=dat_model_pitch_role
)
resultFU2_sm = modelFU2_sm.fit(reml=True, method="lbfgs")
print("Model FU2 sm: CF ~ pitchRole + (1|groupID)")
print(resultFU2_sm.summary())
```

```
Model FU2 sm: CF ~ pitchRole + (1|groupID)
          Mixed Linear Model Regression Results
============================================================
Model:              MixedLM Dependent Variable: cf
No. Observations:   84      Method:             REML
No. Groups:         28      Scale:              1.8873
Min. group size:    3       Log-Likelihood:     -153.6994
Max. group size:    3       Converged:          Yes
Mean group size:    3.0
------------------------------------------------------------
               Coef.  Std.Err.    z    P>|z| [0.025 0.975]
------------------------------------------------------------
Intercept       3.735    0.268 13.937 0.000  3.210  4.260
pitchRole[T.1] -0.581    0.328 -1.774 0.076 -1.224  0.061
groupID Var     0.480    0.268
============================================================


/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-
packages/statsmodels/regression/mixed_linear_model.py:1634: UserWarning: Random
effects covariance is singular
  warnings.warn(msg)
```

2. Bambi (Robustness testing)

[62]:
```
modelFU2_bmb  = bmb.Model(
    "cf ~ pitchRole + (1|groupID)",
    data=dat_model_pitch_role,
    family="gaussian"
)

idata_modelFU2_bmb = modelFU2_bmb.fit(
    draws=2000,
    tune=2000,
    chains=4,
    cores=4,
    random_seed=123
)

print("Model FU2 bmb: CF ~ pitchRole + (1|groupID)")
az.summary(idata_modelFU2_bmb, var_names=["pitchRole", "1|groupID_sigma",␣
 ↪"sigma"])
```

```
Initializing NUTS using jitter+adapt_diag…
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [sigma, Intercept, pitchRole, 1|groupID_sigma, 1|groupID_offset]
```

```
Output()
```

```
Sampling 4 chains for 2_000 tune and 2_000 draw iterations (8_000 + 8_000 draws
total) took 1 seconds.

Model FU2 bmb: CF ~ pitchRole + (1|groupID)
```

[62]:

	mean	sd	eti89_lb	eti89_ub	ess_bulk	ess_tail	r_hat	\
pitchRole[1]	-0.589	0.338	-1.1	-0.053	7342	5596	1.00	
1|groupID_sigma	0.66	0.28	0.17	1.1	1506	1845	1.00	
sigma	1.408	0.137	1.2	1.6	2715	3478	1.00	

	mcse_mean	mcse_sd
pitchRole[1]	0.0039	0.0028
1|groupID_sigma	0.0072	0.005
sigma	0.0026	0.002